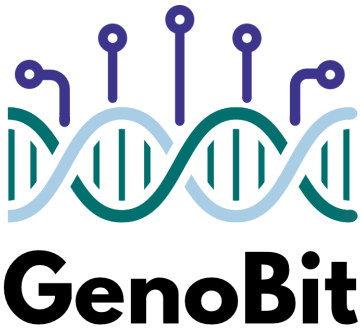



# Neuroimaging

## Etapa 1: Preprocesamiento de los datos

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# 1 Introduction

Alzheimer'z disease (AD) is a neurodegenerative disease in elderly people, that causes memory lost and impared thinking. It is characterized by the build up of two main proteins amyloid beta and tau protein that cause abnormal protein clumps called amyloid plaques and tangled fibers known as neurofibrillary tangles in specific areas of the brain, particularly the temporal lobe and neocortical regions (Dash et al., 2023). There are curently around 50 million people worldwide are living with Alzheimer's, and this number is expected to rise to approximately 152 million by 2050 (Ferreira, C. A. et al., 2018). The progress of AD can be classified as three stages: (i) The asymptotic stage, (ii) Mild AD, and (iii) The Dementia stage. In the first stage, changes in the brain begin to occur without any behavioral symptoms. In the second stage, memory complaints may become noticeable to patients (Mahendran and PM, 2022). In the final dementia stage, they are severely impaired and unable to manage their daily lives. There is no known cure, but if diagnosed in an early asymptomatic stage it can be controled. Recent disease-modifying therapies targeting amyloid-beta plaques have shown modest slowing of cognitive decline in early stages. Lecanemab (Cohen et al., 2023) and donanemab (Courade et al., 2025) are two FDA-approved monoclonal antibodies that reduce amyloid plaque and slow down the progression by approximately 27–35% over 18 months in amyloid-positive patients with mild cognitive impairment or mild AD